

NanoAOD Lepton Reweighting and Validation

CMS Simulation Study

June 20, 2026

Abstract

This note documents a small NanoAOD-level workflow used to distort lepton kinematics and identification efficiencies in simulated events. The modified NanoAOD files are compared to the raw inputs through one-dimensional distributions, efficiency curves, tag-and-probe mass fits, and two-dimensional correlation maps. The figures shown here are produced from the full set of input ROOT files with the current `modify_nanoaod` and `plot.py` configuration.

1 Workflow

The C++ program `modify_nanoaod.cpp` reads NanoAOD `Events` trees, applies deterministic lepton momentum scale and resolution changes, and distorts selected lepton ID/isolation working points according to the configuration in `config.json`. The modified files are written to the `modified/` directory with the configured suffix.

The Python program `plot.py` compares each raw input file to its corresponding modified output. It produces normalized lepton distributions, dilepton spectra, MC-truth and tag-and-probe efficiency curves, fit diagnostic plots, and two-dimensional heatmaps. For every input/output heatmap pair, an additional `Modified - raw` map is produced.

2 Configuration

The lepton momentum scale, resolution, and efficiency distortions use moderate configured amplitudes. The barrel and endcap scale shifts are kept in the same direction with small regional differences, and the efficiency turn-on distortions are limited, so the validation changes remain visible without splitting the dilepton invariant-mass peak into artificial multiple peaks. Branches with explicit energy values are scaled by the same ratio as the corresponding transverse momentum, so the energy response follows the configured momentum distortion without introducing an independent mass-shape change.

For tag-and-probe fits, the nominal invariant-mass fit range is kept broad (55–135 GeV). For bins with an additional high-mass background shape, the fit window can be shortened to 100 GeV, while the signal coefficients are kept normalized over the full mass range. The reported efficiency therefore corresponds to the full signal PDF normalization, not only to the restricted fit window. Each accepted fit must satisfy RooFit status and covariance quality requirements and pass a pass/fail χ^2/ndf validation. The χ^2/ndf values and the fit window are printed directly on each fit PDF.

3 Kinematic Distributions

Figure 1 shows representative lepton-level distributions before and after the NanoAOD modification. The momentum spectra show the overall kinematic response, while the smearing response isolates the stochastic resolution term.

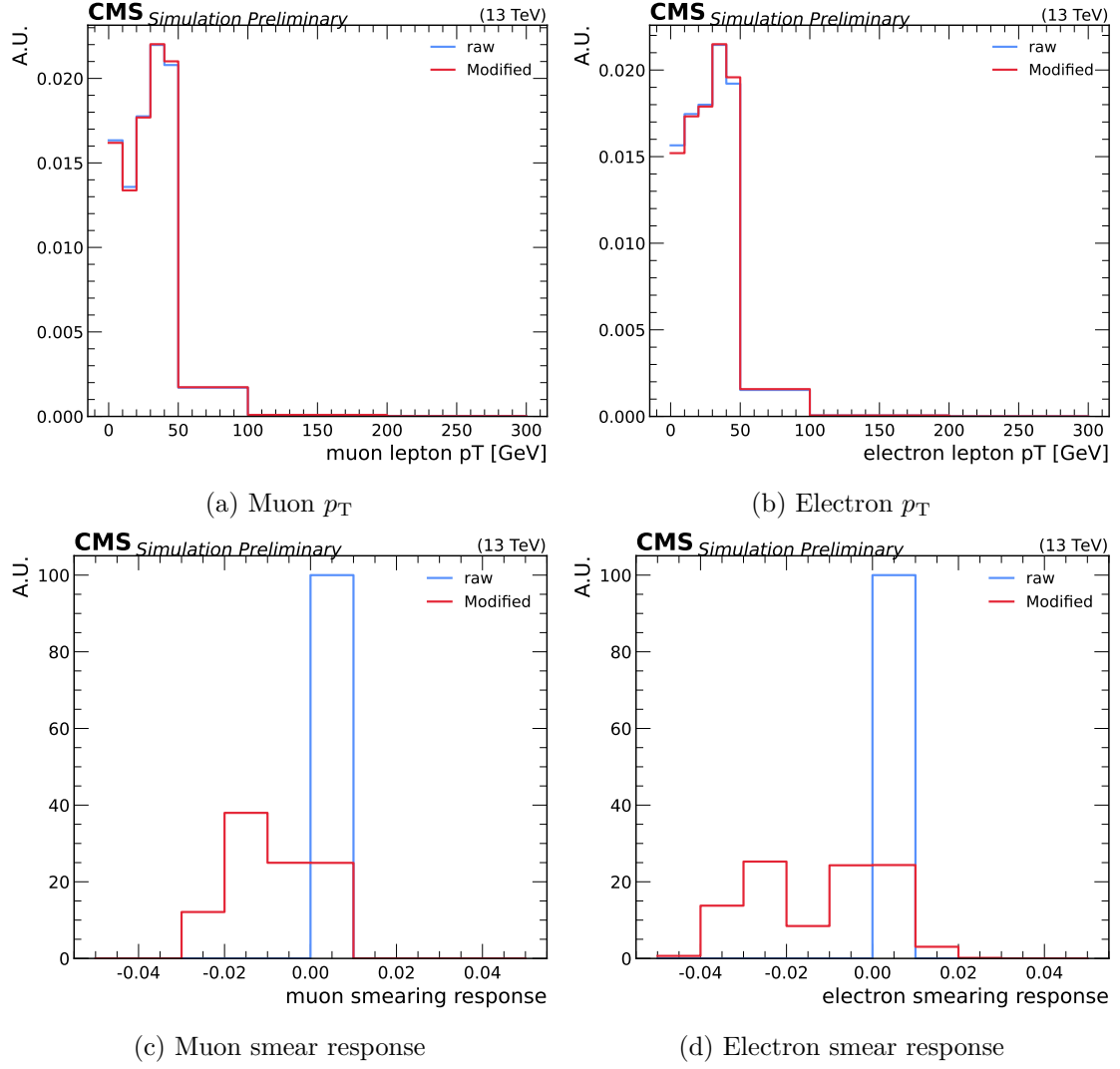


Figure 1: Representative lepton kinematic validation plots.

4 Efficiency Validation

The efficiency validation compares the raw and modified samples with generator-matched MC truth and quality-selected tag-and-probe fits. The modified efficiencies follow the configured global and kinematic distortions. The TnP fit PDFs under `../figures/efficiency/fit/` include the fit range and pass/fail χ^2 /ndf values.

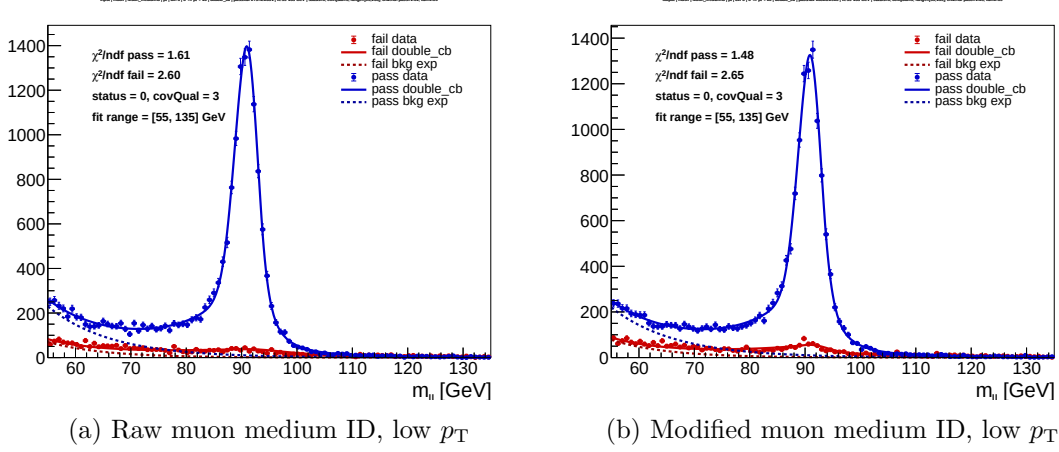


Figure 2: Representative tag-and-probe mass fits.

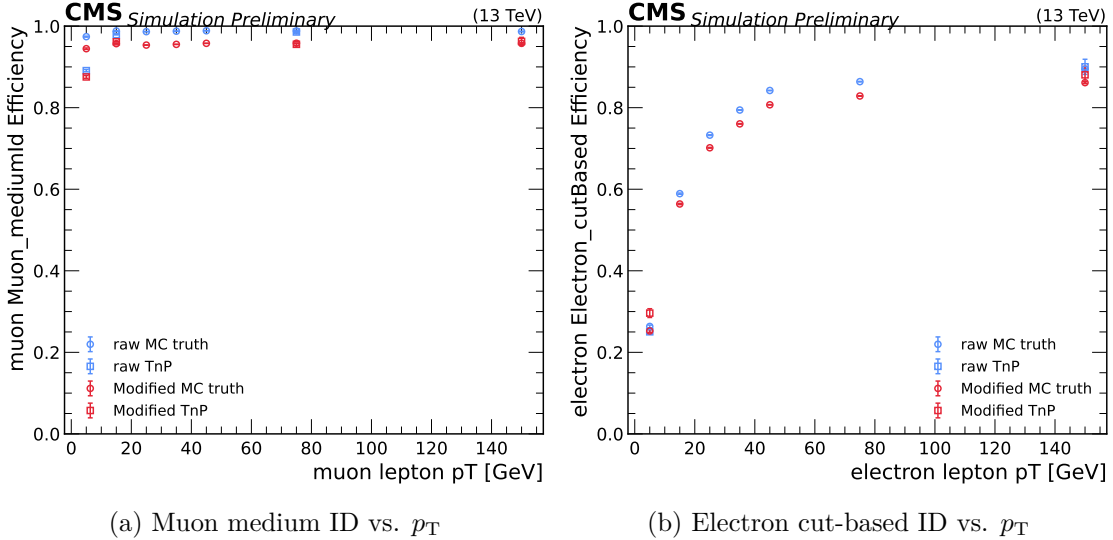


Figure 3: Representative efficiency curves.

5 Dilepton Resolution

The dilepton mass distributions summarize the combined impact of the lepton momentum scale and resolution changes. The modified mass peak can shift or broaden relative to the raw sample according to the configured scale and resolution terms. The validation should therefore check that the change is controlled and does not introduce artificial multi-peak mass shapes.

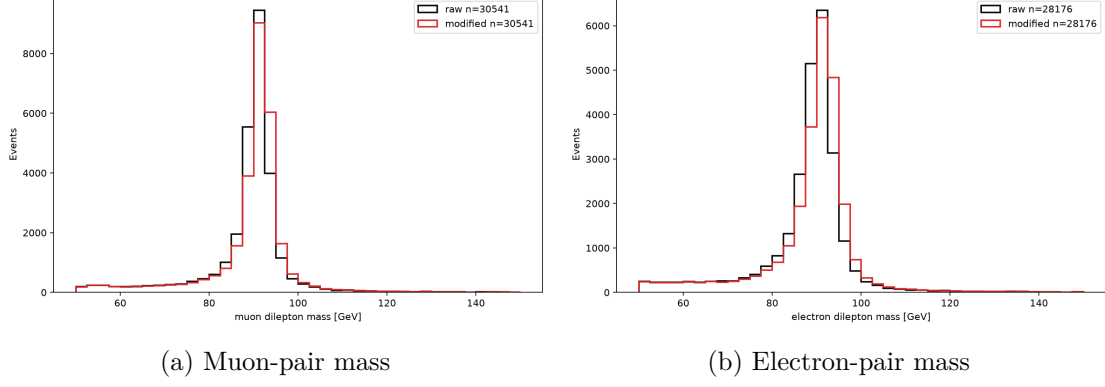


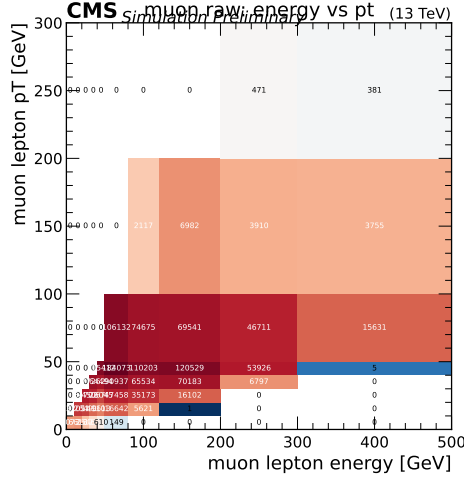
Figure 4: Dilepton mass validation.

6 Two-dimensional Heatmaps

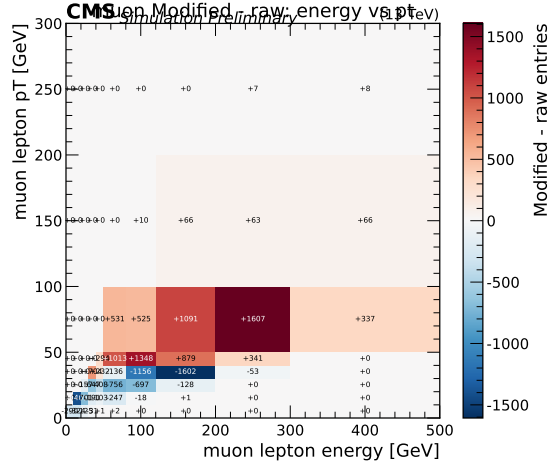
The correlation plots are written in separate `input`, `output`, and `delta` directories. The delta maps show the bin-by-bin `Modified` - `raw` entry difference and provide a compact check of where the modification changes the event population.

7 Production Summary

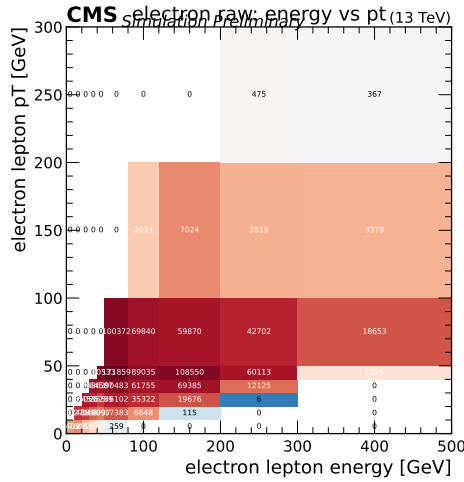
The full production contains seven modified ROOT files in `modified/`. The full plotting step produced 164 PDF figures, including 22 accepted TnP fit diagnostics and 93 correlation heatmaps. No two-peak signal model is used. The modified electron and muon dilepton mass spectra both have a single prominent peak in the 90.0–92.5 GeV bin, consistent with the intended Z-boson validation region.



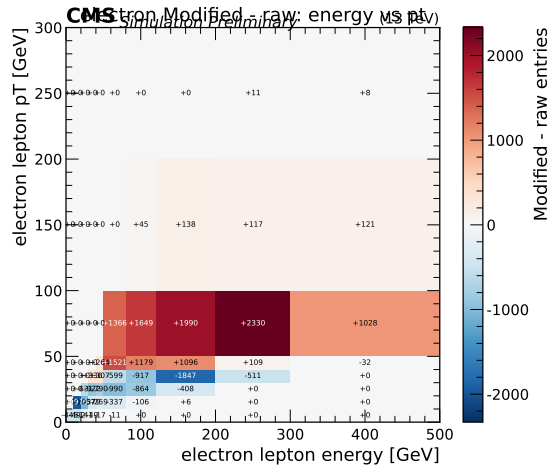
(a) Muon raw energy vs. p_T



(b) Muon modified minus raw



(c) Electron raw energy vs. p_T



(d) Electron modified minus raw

Figure 5: Representative two-dimensional heatmaps.